Table 7.6: Immunization for diphtheria

Immunization		
Active	**Passive**	**Combined**
DPT (Triplevaccine) 3 doses of 0.5ml each are given intramuscular route at an interval of 4–6 weeks after birth. Booster dose of DPT are given at 18 months and at the age of 5.	500–1000 units of Antidiphtheritic serum, (ADS) is administered subcutaneously.	It consists of administration first dose of adsorbed toxoid on one arm, while ADS is given on another arm.

Table 7.7: Clostridium sp. causing pathogenic diseases.

Organisms	Diseases
Clostridium tetani	Tetanus
Clostridium perfringens	Gas gangrene
Clostridium botulinum	Food poisoning

7.7.1 Morphology

They are Gram positive spore forming rods. The spores are spherical and terminal in position giving a drumstick appearance. They are motile and non – capsulated.

7.7.2 Culture Characteristics

- They are obligate anaerobes, optimum temperature is 37°C and pH is 7.4.
- It grows on ordinary media, but growth is enhanced by addition of blood and serum. *Clostridia tetani* grows on the following media and show the characteristic colony morphology (Table 7.8).

7.7.3 Toxins

Clostridium tetani produces two distinct toxins namely,

a. Tetanolysis (haemolysin)

b. Tetanospasmin (neurotoxin)

Table 7.8: Colony characteristics of *Clostridium tetani*

Media	Colony Morphology
Blood agar	They produce α – hemolysis which subsequently develop into β – hemolysis (due to tetanolysis) it produces swarming growth.
Cooked meat broth (CMB)	Growth occurs as turbidity with gas formation. The meat is not digested but becomes black on prolonged incubation

Tetanolysis

- Heat labile and oxygen labile toxin.
- It lysis erythrocytes and also acts as neurotoxin.

Tetanospasmin

- It is heat labile and oxygen stable powerful neurotoxin.
- It is protein in nature. consisting of a large polypeptide chain (93,000 Dalton) and a smaller polypeptide chain (52,000 Dalton) joined by a disulphide bond.
- **Mode of Action:** Tetanospasminis a neurotoxin, which blocks the release of

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